Citation Integrity Audit — Expanded Repository Version

For awesome-tool-selection-agentic-ai | Lab 1 lessons carried into the verified GitHub repository

1. Purpose

This document records the citation-integrity lessons from Lab 1 and prepares them for the GitHub research repository. The purpose is not to rewrite the original Lab 1 paper, but to make the verification process explicit and reusable for the new curated repository.

2. Lab 1 → Lab 2 connection

Lab 1 began with an AI-assisted research paper on evaluating tool-selection accuracy in multi-tool agentic AI systems. The citation audit then tested whether the references were authentic, correctly matched to their metadata, and actually supported the claims for which they were cited. The GitHub repository continues the same topic but raises the verification standard: resources must be independently checked before they are included.

3. Audit outcome

Audit category	Count	Meaning
A — Verified	3	Reference was confirmed and metadata/identity were consistent.
B — Wrong metadata	2	The cited work existed, but one or more bibliographic details were incorrect.
C — Frankenstein	2	The reference combined elements that did not belong to one real publication.
D — Fabricated	1	The claimed reference could not be verified as a real scholarly work.
E — Identifier mismatch	2	An identifier or linked record did not correspond to the claimed work.
Total	10	All ten Lab 1 references were examined.

4. Reported scores

Authenticity score: 55/100.
The audit result demonstrates why an AI-generated bibliography should be treated as a candidate list rather than as verified evidence.

Prediction accuracy: 60%.
This indicates that the audit exercise also evaluated the reliability of the initial judgments against the final verification outcomes.

5. Verification protocol for the GitHub repository

Every scholarly paper added to the repository should pass the following checks:

- Title: the title on the landing page must match the cited title.
- Authors: author names and order should be checked against the authoritative record.
- Year: publication year should agree with the official venue or repository record.
- Venue: journal, conference, workshop, or proceedings information should be consistent.
- Identifier: DOI, ACL Anthology ID, arXiv ID, or another persistent identifier must point to the same work.
- Link target: clicking the link should open the cited paper rather than a different work.
- Relevance: the paper must actually relate to tool use, tool selection, planning, routing, or evaluation.

• Description: the one-line repository description must be based on the actual paper rather than on the title alone.

6. Verification table template

Field	What to record
Paper title	Exact verified title
Authors	Verified authors
Year	Verified year
Venue	Conference/journal/proceedings
Identifier	DOI, arXiv, ACL, or official identifier
Official URL	Landing page used for verification
Category	Survey / foundation / method / evaluation / recent
Relevance	One concise sentence
Verification status	Verified / Needs review

7. Common failure modes

• A real paper title paired with the wrong DOI.

• A DOI that belongs to a different paper.

• A plausible title that cannot be found in an authoritative index.

• A conference/year combination that does not match the publication record.

• A repository link that points to a project rather than the cited paper.

• A description that claims a paper evaluates tool selection when the paper evaluates a broader agent capability.

8. Why this matters for tool-selection research

Tool-selection research is highly interconnected: one paper may study retrieval, another planning, another function calling, and another end-to-end agent evaluation. If bibliographic metadata is wrong, it becomes difficult for another researcher to reproduce the literature review or locate the underlying evidence. The GitHub repository therefore treats verification as part of the research contribution.

9. Final audit checklist

■ At least 20 scholarly papers selected.

■ Each paper has a stable official link.

■ Each paper has author/year/venue information.

■ Each paper has a one-line relevance explanation.

■ Papers are grouped into meaningful categories.

■ No fabricated references are included.

■ No copyrighted papers are uploaded unless redistribution is clearly permitted.

■ The repository clearly distinguishes original Lab 1 material from newly curated resources.

10. Recommended citation-audit appendix

For each future paper, keep a compact audit record. This makes the repository maintainable and makes later corrections easy.

Check	Pass condition	Result
Identity	Title and authors match official record	■
Year	Year matches official record	■
Venue	Venue matches official record	■
Identifier	DOI/ID resolves to same work	■
URL	Official landing page is reachable	■
Relevance	Direct connection to topic	■
Description	Description supported by paper	■

11. Repository policy

The repository should state a simple policy: AI may assist discovery and organization, but the final inclusion decision is made after human verification. This directly addresses the central lesson of Lab 1.

12. Note for submission

This audit document is a prepared expanded companion for the GitHub repository. The original Lab 1 PDF remains the authoritative submitted paper/audit source. If the instructor expects the exact original audit document, upload the original Lab 1 file rather than replacing it with a generated summary.